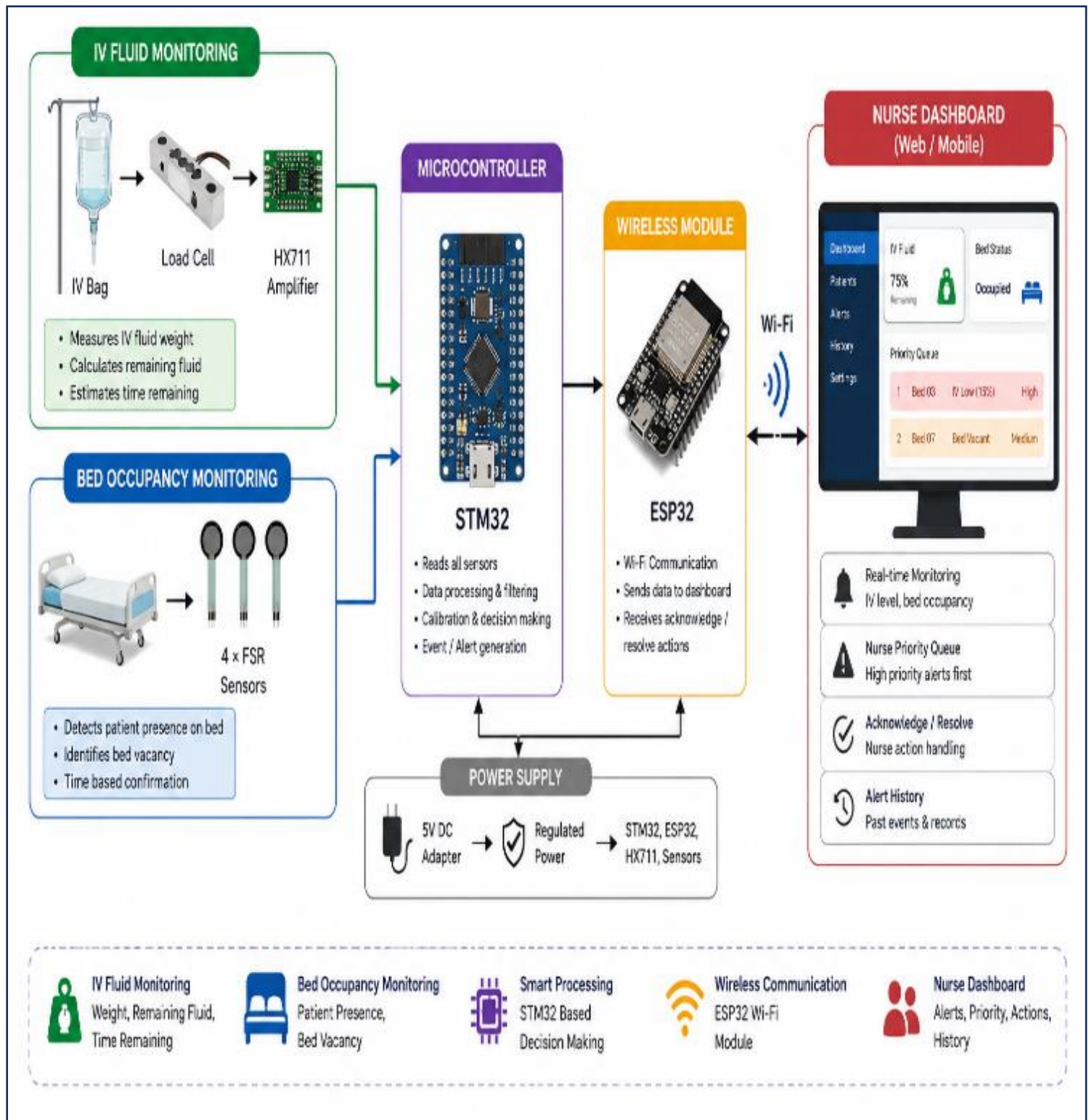
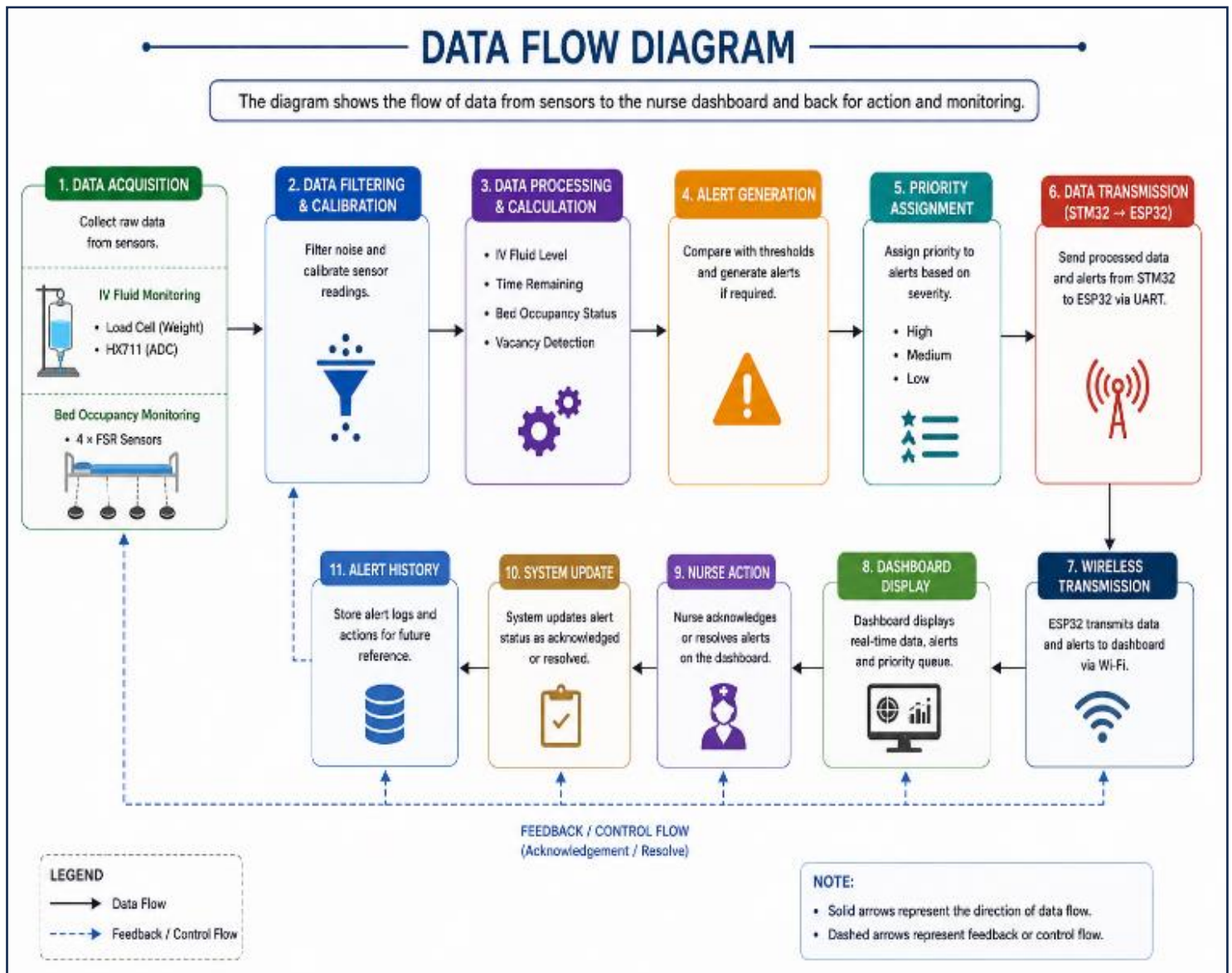


SYSTEM ARCHITECTURE:

Overall Solution Architecture: Shows the major components of the proposed system and their interaction from sensing to nurse notification.



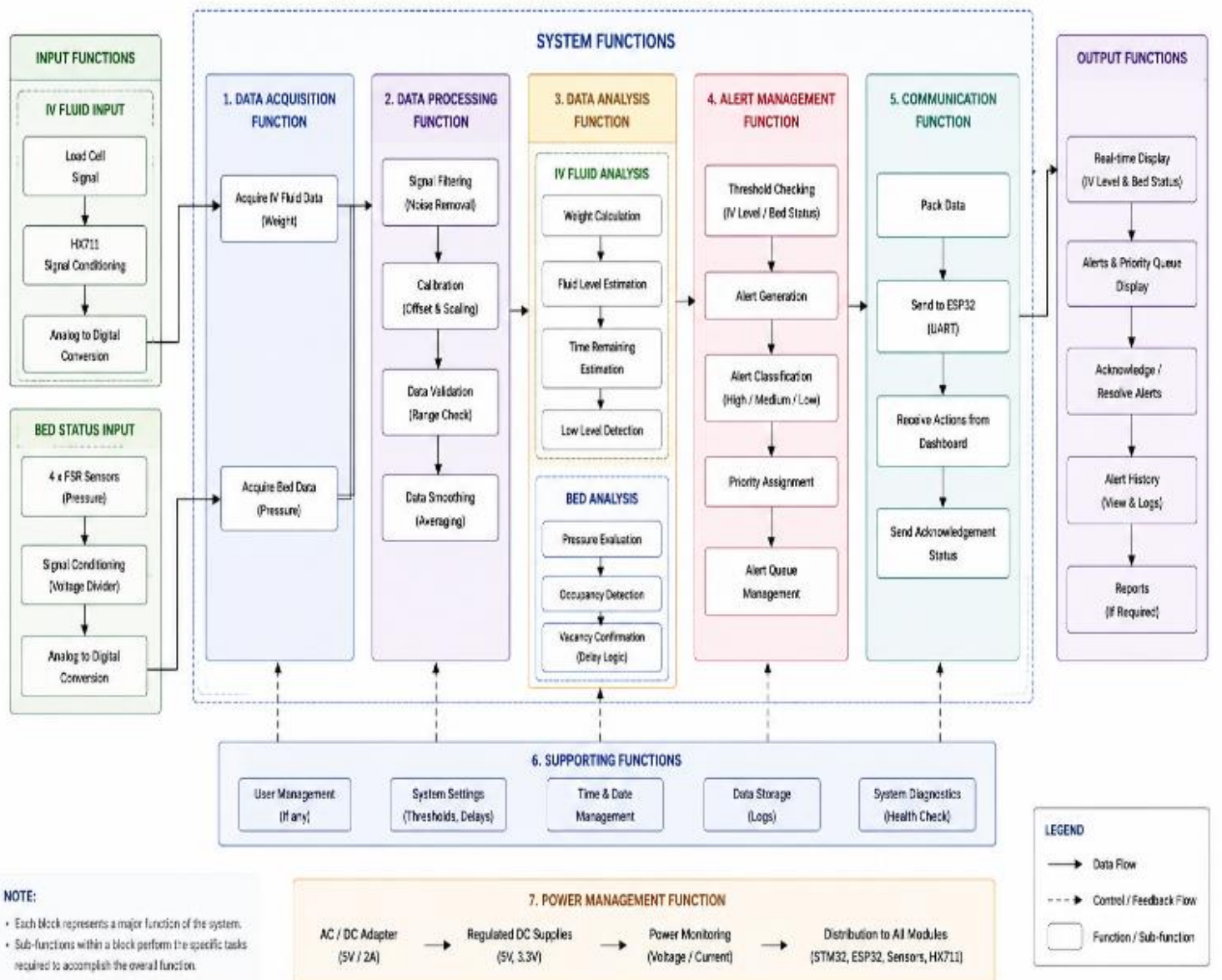
Data Flow Diagram: Shows how sensor data moves from the sensing units through the STM32 and ESP32 to the nurse dashboard.



Functional Block Diagram:

Shows the functional operations performed by each hardware/software block and their connections.

FUNCTIONAL BLOCK DIAGRAM

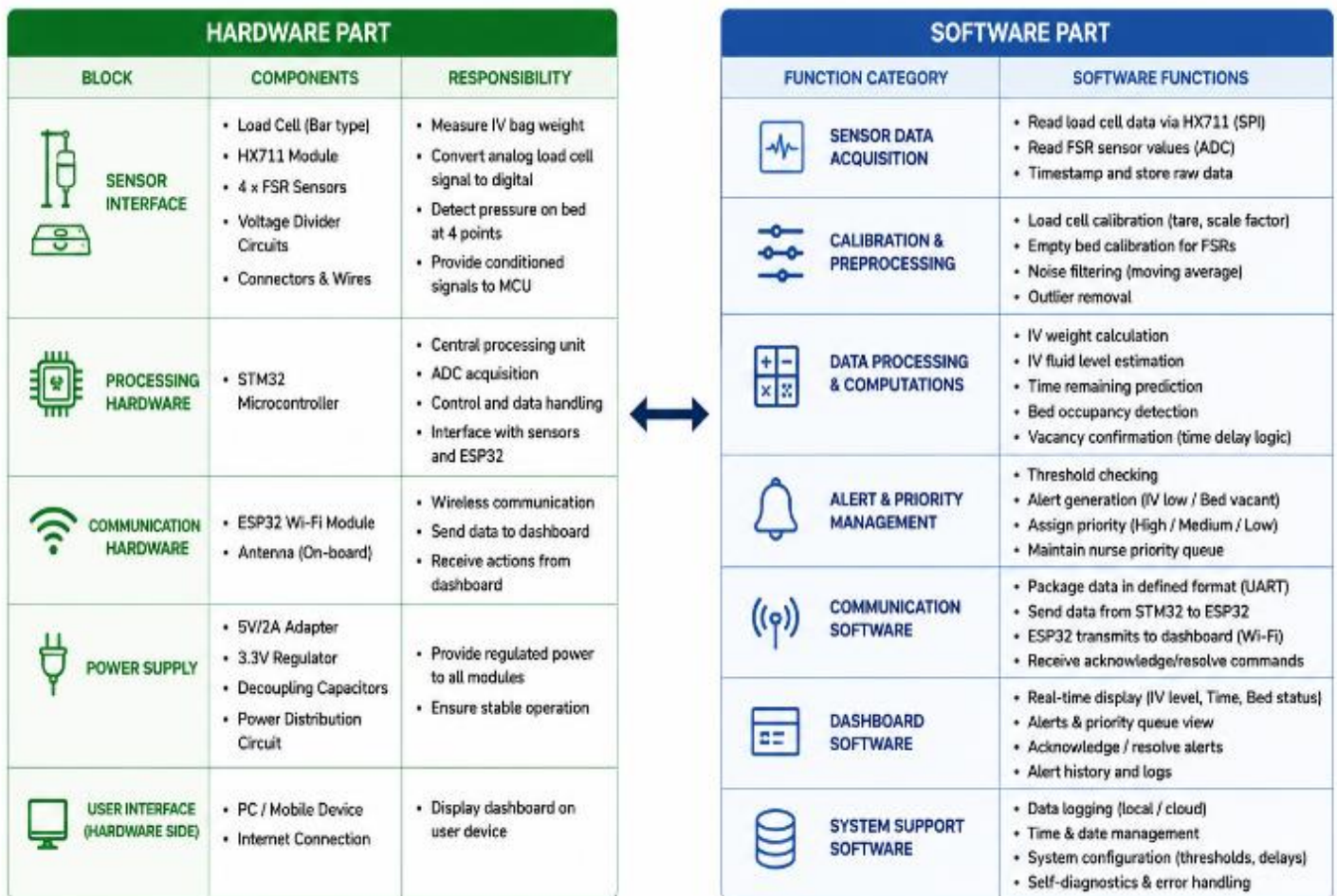


Hardware–Software Partitioning:

Shows which functions are implemented using hardware components and which are handled by software.

HARDWARE – SOFTWARE PARTITIONING

The system is divided into two main parts: the physical hardware components and the software/firmware functions.



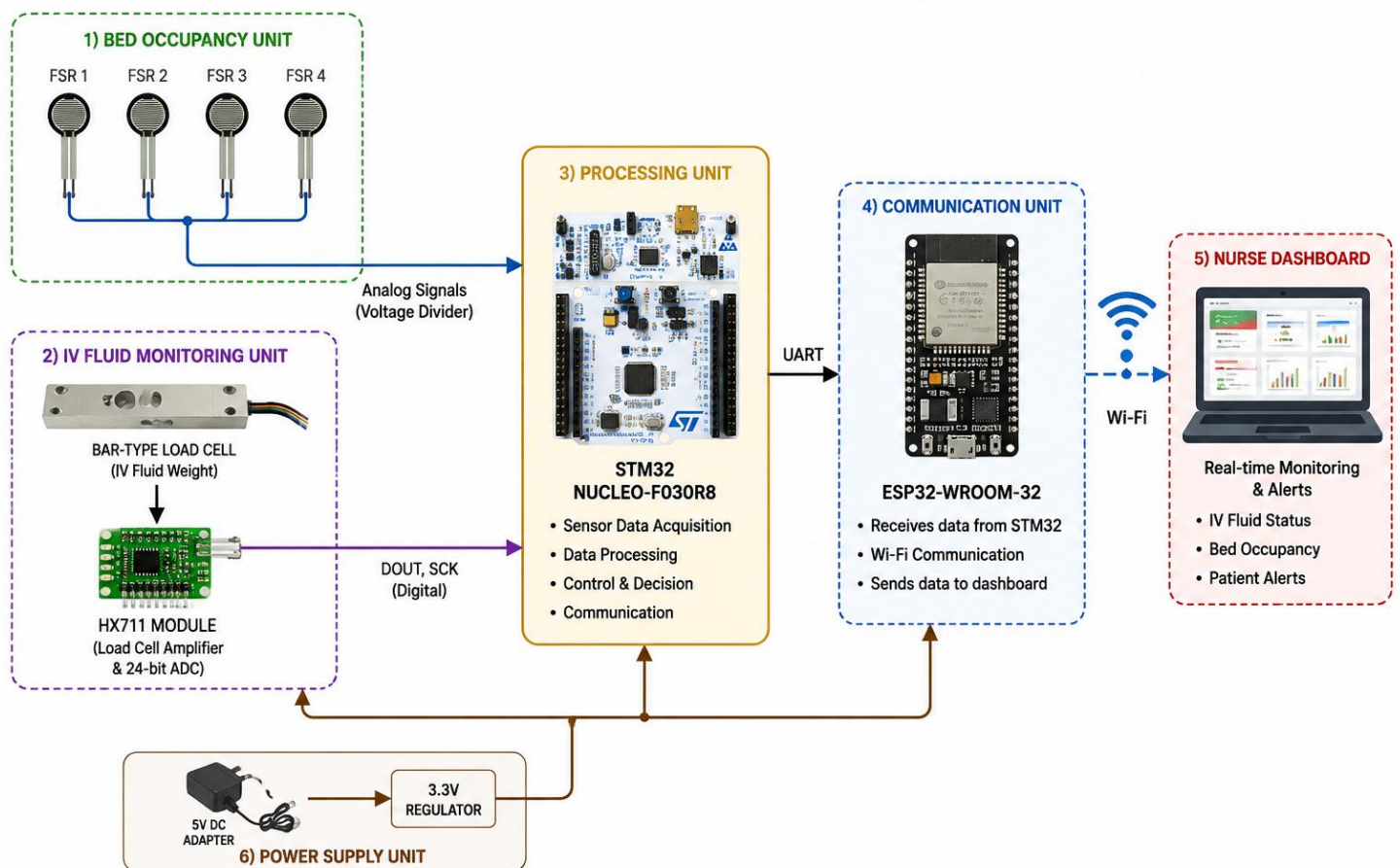
INTERFACE SUMMARY Load Cell → HX711 : Differential Analog HX711 → STM32 : SPI FSR Sensors → STM32 : Analog (ADC) STM32 → ESP32 : UART ESP32 → Dashboard : Wi-Fi (TCP/IP)

HARDWARE DESIGN:

FINAL BLOCK DIAGRAM:

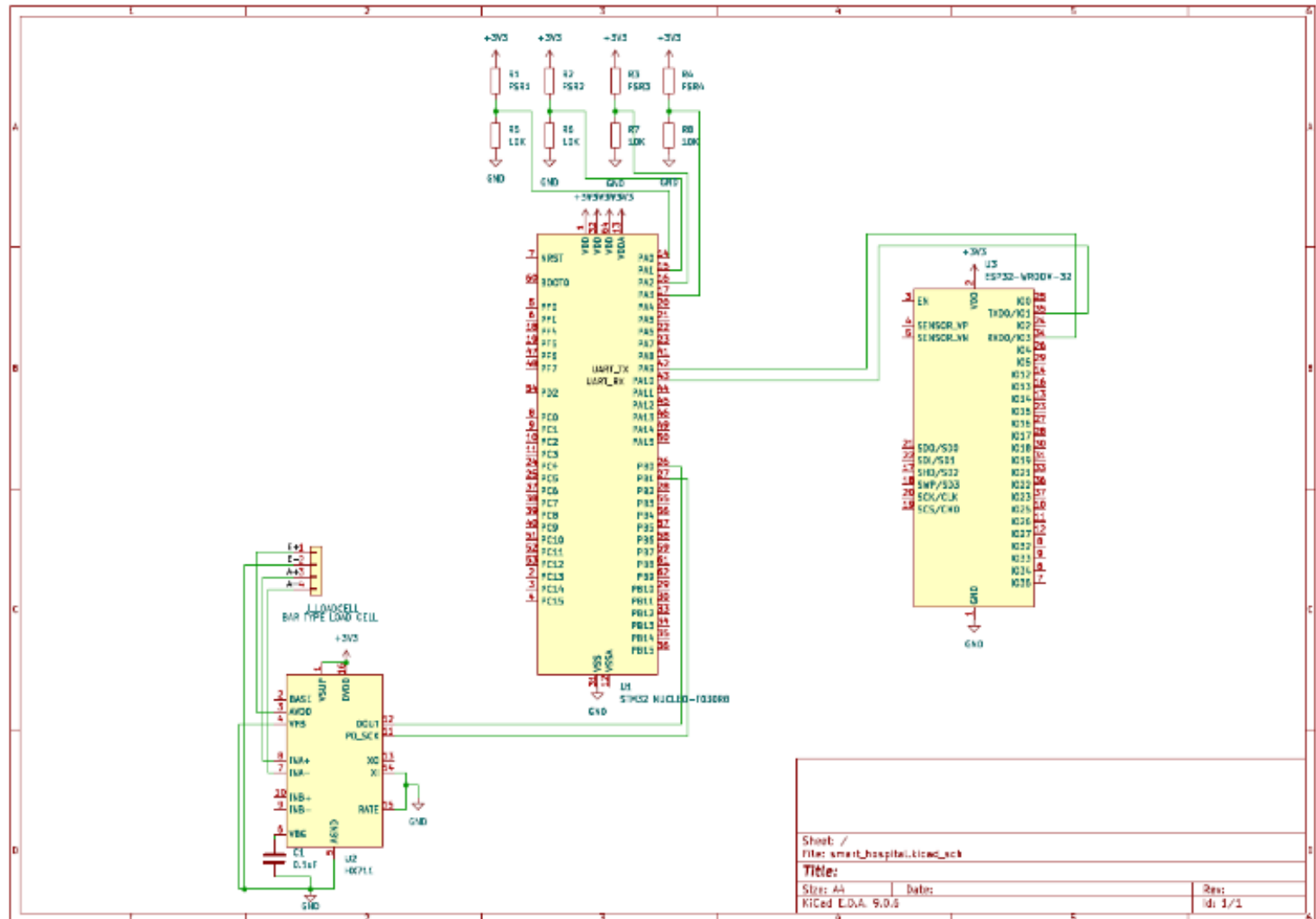
The block diagram shows the major hardware components and their interconnections for bed occupancy and IV fluid monitoring.

FINAL HARDWARE BLOCK DIAGRAM



CIRCUIT SCHEMATICS:

The schematic shows the electrical connections between the FSR sensors, load cell, HX711, STM32 Nucleo, and ESP32.



COMPONENT SELECTION:

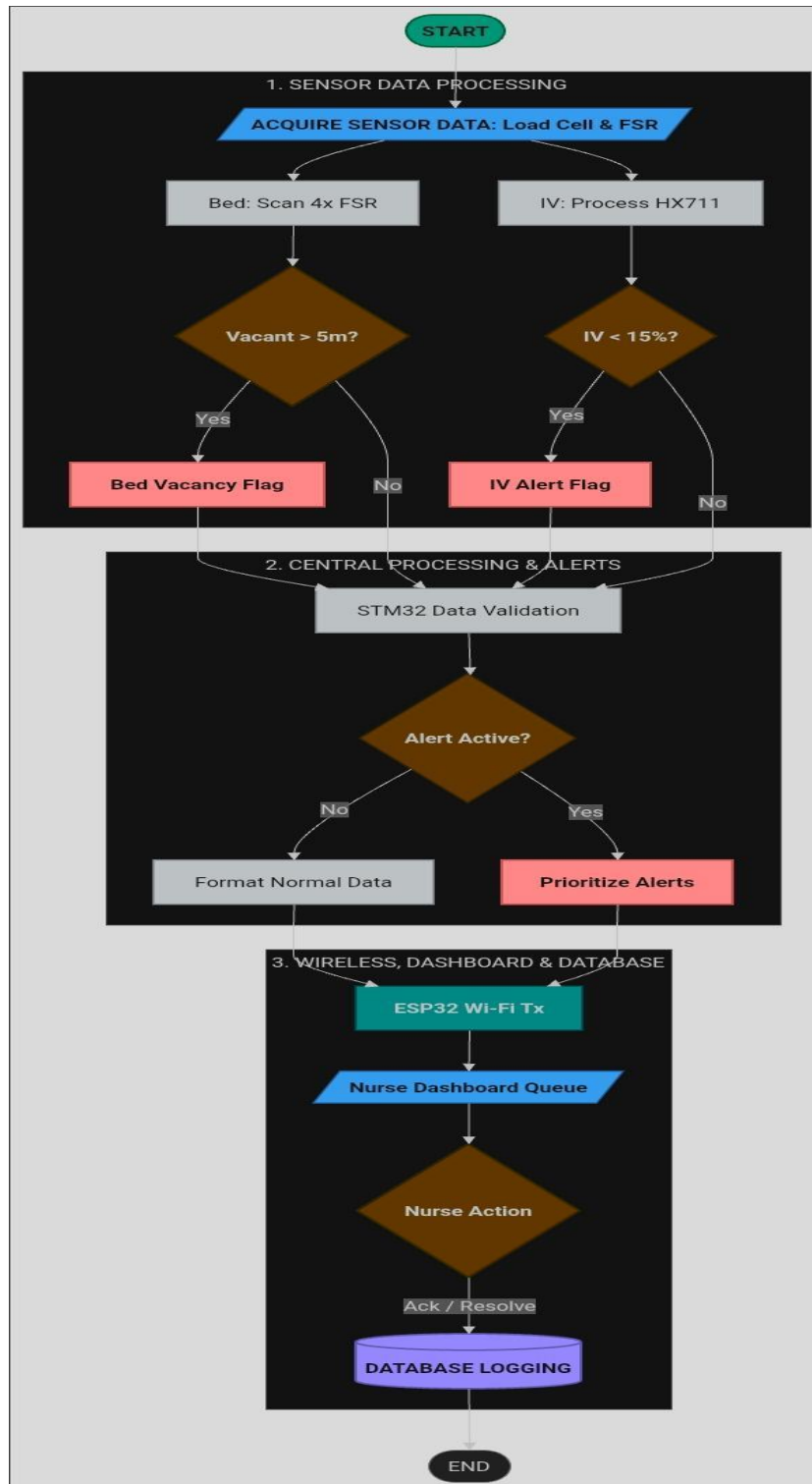
Component	Why needed
STM32 NUCLEO-F030R8	Processes sensor readings
4 FSR Sensors	Detects patient presence on the bed
Bar-type Load Cell	Measures IV fluid weight
HX711	Amplifies and converts load-cell signals
ESP32-WROOM-32	Sends data through Wi-Fi
10 k Ω Resistors	Forms voltage dividers with FSRs
3.3 V Power Supply	Powers the electronic components
0.1 μ F Capacitors	Reduces power-supply noise

BILL OF MATERIALS:

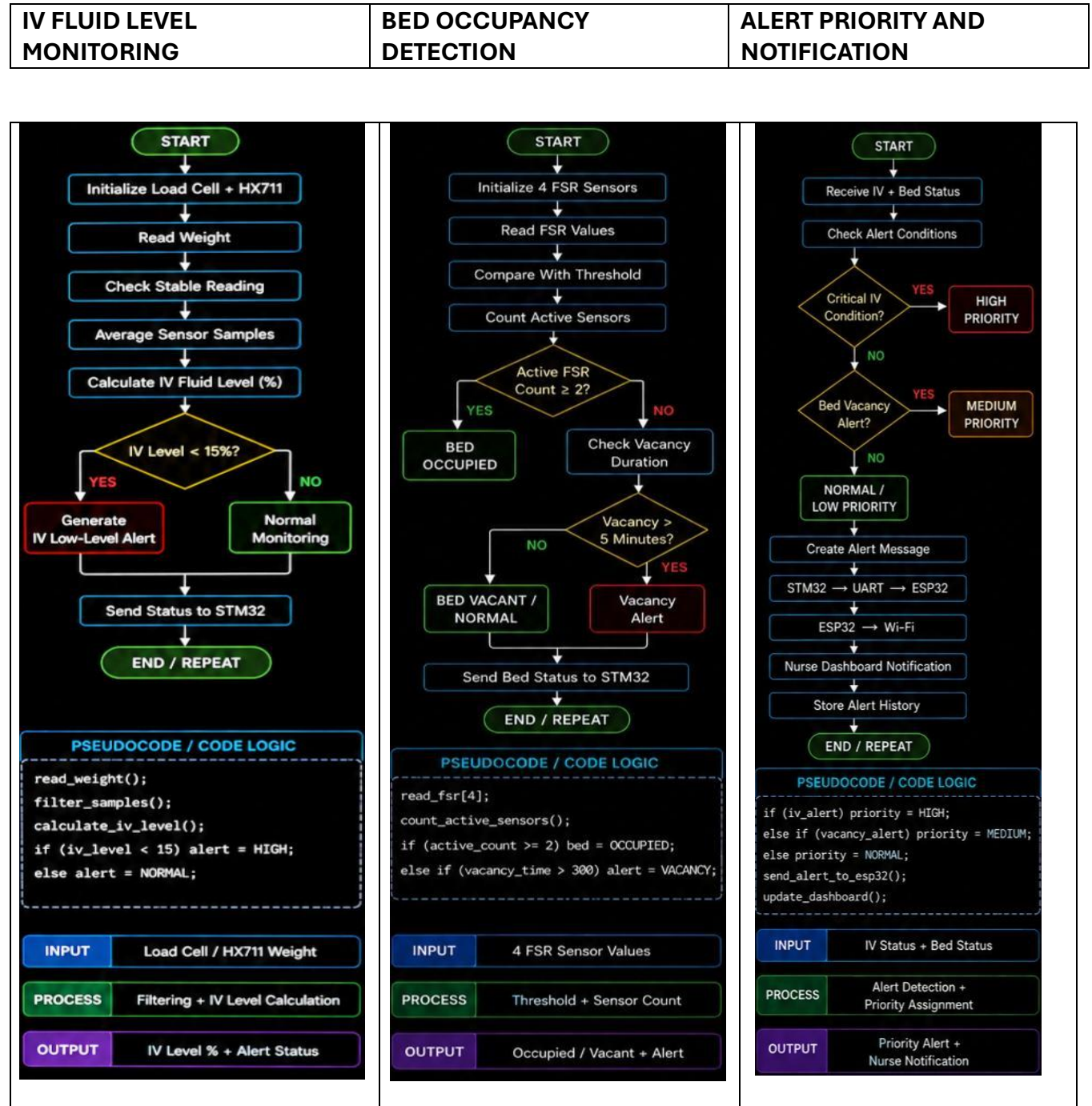
S.No.	Component	Specification	Quantity
1	STM32 NUCLEO-F030R8	STM32F030R8T6, 32-bit MCU	1
2	FSR Sensor	Force Sensitive Resistor	4
3	Bar-type Load Cell	Weight measurement	1
4	HX711 Module	24-bit ADC / Load-cell amplifier	1
5	ESP32-WROOM-32	Wi-Fi module	1
6	Resistor	10 k Ω	4
7	Capacitor	0.1 μ F	As required
8	Connecting Wires	Jumper wires	As required
9	Breadboard/PCB	Prototype assembly	1

SOFTWARE DESIGN:

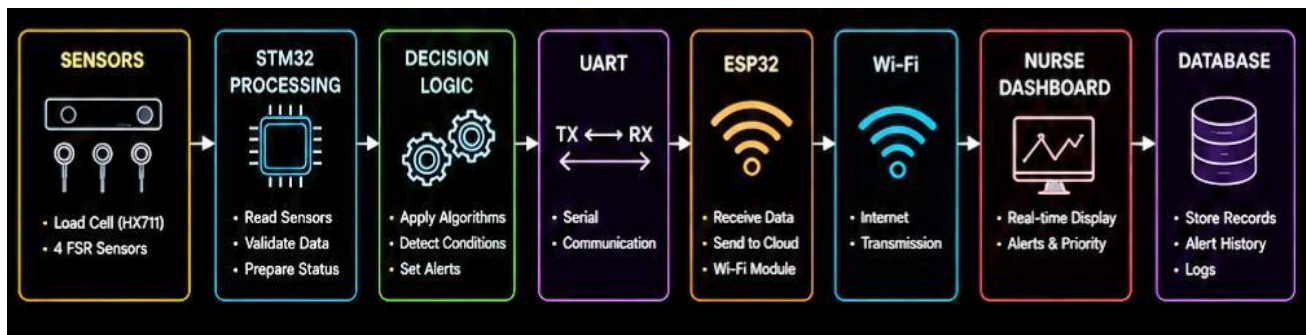
1.FLOW CHART:



2.ALGORITHM :

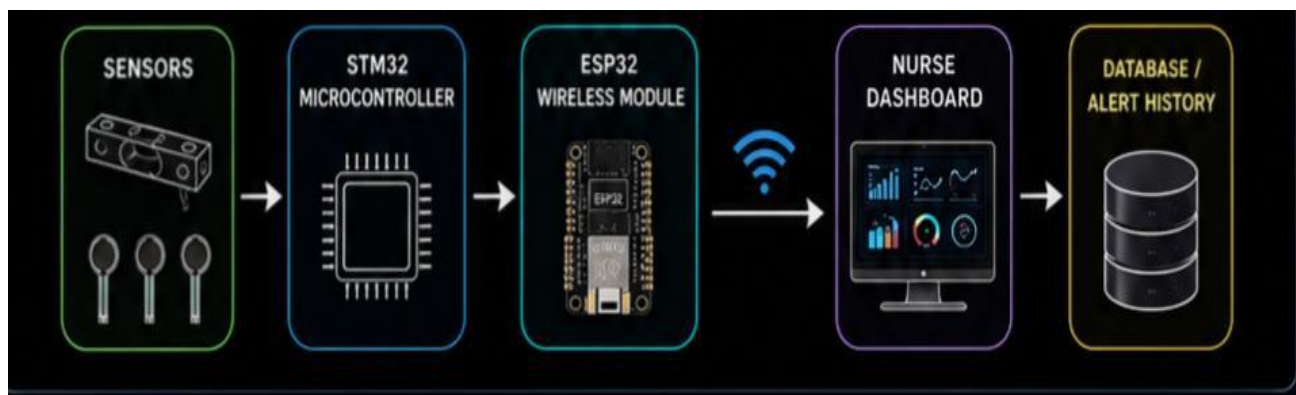


SOFTWARE PIPELINE:



3.COMMUNICATION PROTOCOL:

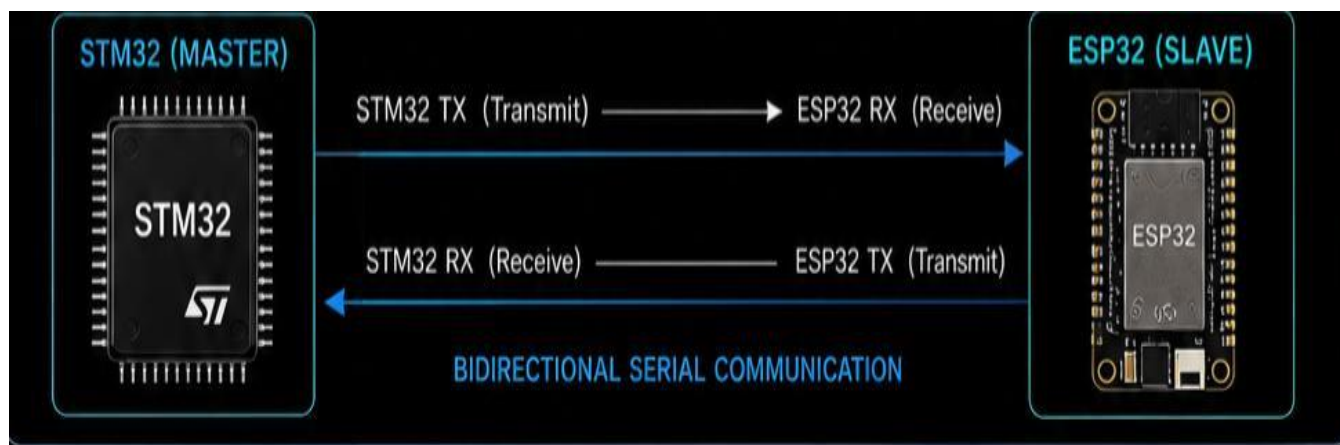
➤ SYSTEM COMMUNICATION ARCHITECTURE



COMMUNICATION LINKS:

Sensor → STM32	GPIO/ADC/ HX711 Interface: Sensors send raw data to STM32 for acquisition.
STM32 → ESP32	UART STM32 transmits processed data packets to ESP32 via UART.
ESP32 → Dashboard	Wi-Fi: ESP32 sends data to the dashboard using Wi-Fi connection.
Dashboard → Database	Internet / API Dashboard stores data to the database using Internet / API.

➤ **UART COMMUNICATION :**



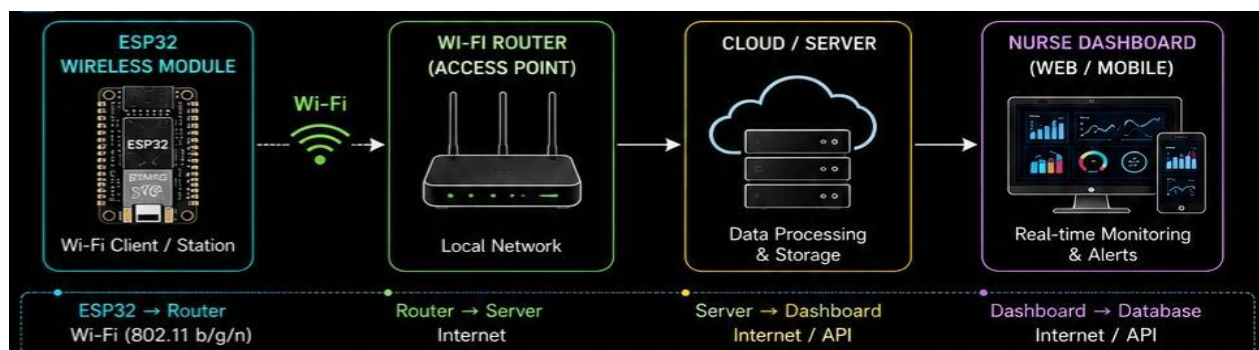
UART SPECIFICATION:

Protocol : UART(Asynchronous serial)
Purpose : STM32-ESP32 data transfer
Data : IV level, bed status, alert priority
Direction : Bidirectional (TX/RX)
Baud Rate : 115200 bps(configurable)
Data Bits : 8
Stop Bits : 1
Parity : None
Flow Control : None
Voltage Level : 3.3V TTL

UART DATA PACKET FORMAT:



➤ **WI-FI COMMUNICATION ARCHITECTURE:**



WI-FI COMMUNICATION DETAILS

Protocol	WI-FI (IEEE 802.11 b/g/n)
Purpose	Wireless monitoring and alert transmission
Data	Real-time Sensor status and alerts
Network Mode	Station (Client Mode)
Frequency Band	2.4 GHz
Security	WPA2-PSK
Data Rate	Up to 72 Mbps
Topology	Infrastructure Mode
IP Address	DHCP/Static(Configurable)
Transport	TCP/UDP(MQTT/HTTP)

➤ **FINAL COMMUNICATION SEQUENCE:**

